Lecture 7: The Simplex Method

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Feb 15, 2023

Announcement

- Homework 1 due today at noon.
- Homework 2 posted, due next Wednesday (2/22), at noon.
- No class on Friday (2/17).

Recap

In the previous lectures, we showed an important fact about linear optimization

- If there is an optimal solution, then there is an optimal basic feasible solution
- Therefore, when we solve LP, we only need to search among the BFS

We introduced the simplex method, which is a *smart* way to search among the BFS

- It starts from a BFS
- Then it searches among its neighbors (differ by only one index among the basic indices)
- Either one can find a better neighbor or one knows that the current solution is already optimal

Simplex Method

When at a BFS, the simplex method considers its neighbors by choosing one entering basis.

- When the incoming index is chosen, the (jth feasible) direction is also set:

$$\mathbf{d} = [-A_B^{-1}A_j; 0; ..; 1; ..; 0]$$

- The change to the objective value when going in jth feasible direction (in a unit step size) is given by the reduced cost $\bar{c}_j$

$$\bar{c}_j = c_j - \mathbf{c}_B^T A_B^{-1} A_j$$

- When all the reduced costs are non-negative, the current BFS is optimal

Change of Basis

What if some $\bar{c}_j < 0$?

- Then it means that by bringing jth variable (non-basic) into the basis, we can decrease the objective value. Thus we want to go in that direction

We calculate the stepsize:

$$\theta^* = \min_{\{i \in B | d_i < 0\}} \left(-\frac{x_i}{d_i} \right)$$

The new solution (next BFS) is

$$\mathbf{y} = \mathbf{x} + \theta^* \mathbf{d}$$

And the new basis is

$$B(1), ..., B(\ell - 1), j, B(\ell + 1), ..., B(m)$$

where $B(\ell) \in \{B(1), ..., B(m)\}$ is the index such that $\theta^* = -\frac{x_{B(\ell)}}{d_{B(\ell)}}$

Change of Basis

Why $\mathbf{y}$ is a new BFS? Obviously, $\mathbf{y} \geq 0$, and $A\mathbf{y} = \mathbf{b}$, and also $\mathbf{y}$ only contains positive elements within the columns:

$$B(1), ..., B(\ell - 1), j, B(\ell + 1), ..., B(m)$$

We only need to argue that these columns must be linearly independent (which is required for basic solutions).

We prove by contradiction. If they are not linearly independent, it must be that column j can be represented as linear combinations of columns $B(1), ..., B(\ell - 1), B(\ell + 1), ..., B(m)$. Then when consider $A_B^{-1} A_j$, the entry corresponding to $B(\ell)$ must be zero. That means $d_{B(\ell)}$ is 0.

However, remember $\theta^* = \min_{\{i \in B | d_i < 0\}} \left(-\frac{x_i}{d_i} \right)$. If $d_{B(\ell)} = 0$, then it cannot be that $B(\ell)$ is the index to attain the minimal ratio, which also means that it cannot be $B(\ell)$ that leaves the basis. $\square$

An Iteration of the Simplex Method

We start from a BFS $\mathbf{x}$ (with corresponding basis B)

1. We first compute the *reduced costs* $\bar{\mathbf{c}}$

$$\bar{c}_j = c_j - \mathbf{c}_B^T A_B^{-1} A_j$$

   - If no reduced costs is negative, then $\mathbf{x}$ is already optimal
   - Otherwise choose some j such that $\bar{c}_j < 0$

2. Compute the jth basic direction $\mathbf{d} = [-A_B^{-1}A_j; 0; ...; 1; ...; 0]$
   - If $\mathbf{d} \geq 0$, then the problem is unbounded, i.e., the optimal value is $-\infty$.
   - Otherwise, compute $\theta^* = \min_{i \in B, d_i < 0} \{-\frac{x_i}{d_i}\}$

3. Let $\mathbf{y} = \mathbf{x} + \theta^* \mathbf{d}$. Then $\mathbf{y}$ is the new BFS with index j replacing $B(\ell)$ in the basis, where $B(\ell)$ is the index attaining the minimum in θ^*. Objective value is changed by $\theta^* \mathbf{c}^T \mathbf{d} = \theta^* \bar{c}_j$.

4. Repeat these procedures

Degeneracy

In most of the cases, the objective value will strictly decrease after one simplex method iteration. However, it is possible that the objective stays the same.

Since the change of the objective value (if one chooses to have x_j enter the basis) is $\theta^* \bar{c}_j$ and we know that $\bar{c}_j < 0$. This can only happen if $\theta^* = 0$.

Recall that

$$\theta^* = \min_{\{i \in B | d_i < 0\}} \left(-\frac{x_i}{d_i} \right)$$

If for i's such that $d_i < 0$, there exists $x_i = 0$, then $\theta^* = 0$. This may happen when there are 0s in the BFS $\mathbf{x}$.

Degeneracy

Definition (Degeneracy)

We call a basic feasible solution $\mathbf{x}$ degenerate if some of the basic variables are 0.

- Degeneracy could happen. As an algorithm, we need to consider what consequences it may have

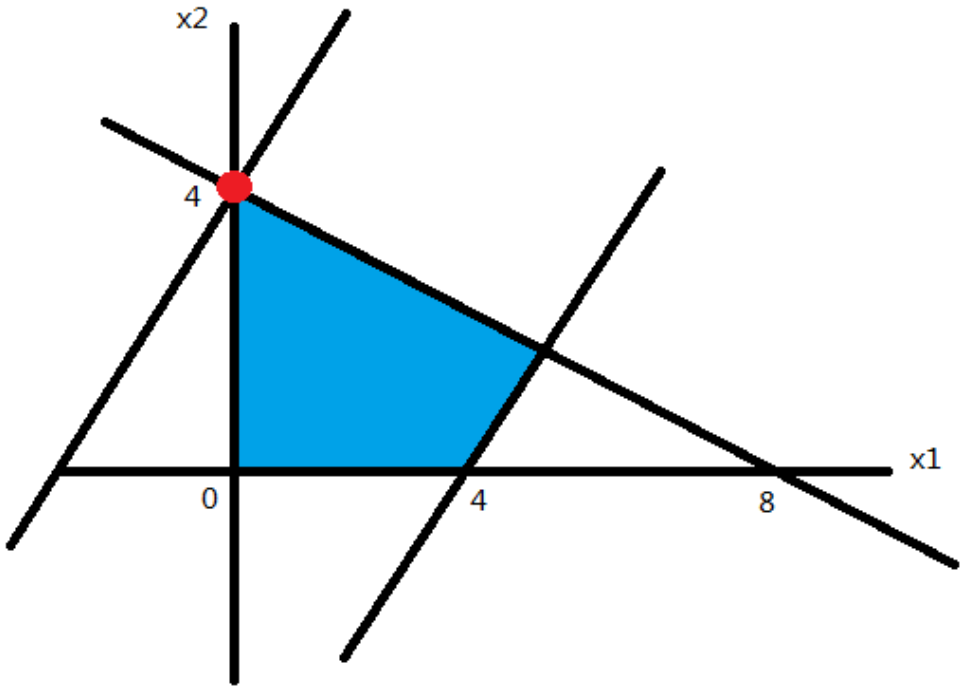
An example:

$$\begin{array}{rrrrrcl} x_1 & +2x_2 & +x_3 & & & = & 8 \\ x_1 & -x_2 & & +x_4 & & = & 4 \\ -x_1 & +x_2 & & & +x_5 & = & 4 \\ x_1 & , x_2 & , x_3 & , x_4 & , x_5 & \geq & 0 \end{array}$$

If we choose the basic indices to be $\{1, 2, 4\}$, then the corresponding basic solution is $(0, 4, 8)$. It is therefore degenerate.

- The is equivalent to that the number of non-zeros at a basic solution is strictly less than m

Illustration



- More than 2 lines intersect at one point
- In general, at degeneracy point, more than k planes intersect at k-dimensional space.

Degeneracy

Assume degeneracy happens at some point:

- Given a BFS $\mathbf{x}$ with negative reduced cost $\bar{c}_j < 0$ and $\theta^* = 0$. And i is the index that achieves $\min_{\{i \in B, d_i < 0\}} (-x_i/d_i)$. Thus, $x_i = 0$.

We can still view that we changed the basis from i (i leaving the basis) to j (j entering the basis) and proceed to the next iteration.

- Although the solution (and the objective value) does not change, the basis changed. Therefore, the reduced costs vector will change in the next iteration — issue seems resolved

However, we need to guarantee that there won't be any cycle, i.e., we will not visit the same BFS more than once

- This can only happen together with degeneracy, since otherwise the objective value will strictly decrease

Example of Cycling

If not dealt properly, cycle can happen. Consider the following LP:

$$A = \begin{pmatrix} -2 & -9 & 1 & 9 & 1 & 0 \\ 1/3 & 1 & -1/3 & -2 & 0 & 1 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$
$$\mathbf{c} = (-2, -3, 1, 12, 0, 0)$$

If we set $B = \{5, 6\}$ initially, then the sequence shown below leads to a cycle (objective value doesn't change, and there is always an index with negative reduced cost):

Step #	1	2	3	4	5	6
Exiting	x_6	x_5	x_2	x_1	x_4	x_3
Entering	x_2	x_1	x_4	x_3	x_6	x_5
Basis	$(5, 2)$	$(1, 2)$	$(1, 4)$	$(3, 4)$	$(3, 6)$	$(5, 6)$

We will show that cycle can be avoided by designing how to choose incoming/outgoing basis when there are multiple choices.

Pivoting Rules: Choose the Entering Basis

In the description of the algorithm, we say that at each feasible solution, we can choose *any* j with negative reduced cost to enter the basis in the next iteration. Sometimes, there are more than one j with $\bar{c}_j < 0$. In this case, we need to make some rules to choose the entering basis.

Here are several possible rules:

1. *Smallest subscript rule*: choose the smallest index j such that $\bar{c}_j < 0$
2. *Most negative rule*: choose the smallest $\bar{c}_j$
3. *Most decrement rule*: choose j with the smallest $\theta^* \bar{c}_j$

Pivoting Rules: Choose the Exiting Basis

Recall that

$$\theta^* = \min_{\{i \in B | d_i < 0\}} \left(-\frac{x_i}{d_i} \right)$$

And we choose the index that attains this minimum to leave the basis. It is possible that there are two or more indices that attain the minimum (tie). Then we also need a rule to decide the outgoing basis.

- The most commonly used rule is the *smallest index rule*

When this tie happens, the next BFS will be degenerate

Bland's Rule

Theorem (Bland's Rule)

If we use both the smallest index rule for choosing the entering basis and the exiting basis, then no cycle will occur in the simplex algorithm.

- The proof is given in Bertsimas's Book

Using the Bland's rule when applying the simplex method, we can guarantee to stop within a finite number of iterations at an optimal solution.

Finding an Initial BFS

In our previous discussion, we assumed that we start with a certain BFS

- This can be done easily if the standard form is derived by adding slacks to each constraint and the right hand side is all nonnegative. (Why?)

However, in general, it is not necessarily easy to get an initial BFS from the standard form. For example,

$$
\begin{array}{lrrrrcr}
\text{minimize} & x_1 & +x_2 & +x_3 & & & \\
\text{subject to} & x_1 & +2x_2 & +3x_3 & & = & 3 \\
 & & -4x_2 & -9x_3 & & = & -5 \\
 & & & +3x_3 & +x_4 & = & 1 \\
 & x_1 & , x_2 & , x_3 & , x_4 & \geq & 0
\end{array}
$$

Finding an Initial BFS

- One could test different basis B, to see if $A_B^{-1}\mathbf{b} \geq 0$.
- However, this may take a long time.
- In fact, in terms of computational complexity (which we will define later), finding one BFS is as hard as finding the optimal solution!

We will discuss an initialization method next — two-phase method.

Two-Phase Simplex Method

In the two-phase simplex method, we first solve an auxiliary problem ($\mathbf{e}$ means an all-one vector).

$$\begin{aligned} \text{minimize}_{\mathbf{x},\mathbf{y}} \quad & \mathbf{e}^T\mathbf{y} \\ \text{subject to} \quad & A\mathbf{x} + \mathbf{y} = \mathbf{b} \\ & \mathbf{x}, \mathbf{y} \geq 0 \end{aligned}$$

Without loss of generality, we assume $\mathbf{b} \geq 0$ (otherwise, we pre-multiply that row by -1).

There is a trivial BFS to the auxiliary problem: $(\mathbf{x} = 0, \mathbf{y} = \mathbf{b} \geq 0)$ so one can apply the Simplex method to solve it.

Theorem

The original problem is feasible if and only if the optimal value of the auxiliary problem is 0.

Proof of the Theorem. First, if the original problem is feasible with a feasible solution $\mathbf{x}_0$. Then $\mathbf{x} = \mathbf{x}_0$, $\mathbf{y} = 0$ is a feasible solution to the auxiliary problem with objective value 0. Note that the optimal value of the auxiliary problem cannot be less than 0. Therefore the optimal value of the auxiliary problem must be 0. ("only if" part proved)

Second, if the optimal value of the auxiliary problem is 0. Say $(\mathbf{x}^*, \mathbf{y}^*)$ is the optimal solution. Then it must be that $\mathbf{y}^* = 0$. Then $\mathbf{x}^*$ is a feasible solution to the original problem. □

Two-Phase Simplex Method

By this theorem, we can solve the auxiliary problem by the Simplex method, and

1. If the optimal value is not 0, then we can claim that the original problem is infeasible;
2. If the optimal value is 0 with solution $(\mathbf{x}^*, \mathbf{0})$. Then we know that $\mathbf{x}^*$ must be a BFS for the auxiliary problem. Then it must be a BFS for the original problem as well. And we can start from there to initialize the simplex method.

If $\mathbf{x}^*$ is degenerate (has less then m positive entries) and still contains basic indices in the auxiliary variables, then one can pick any other columns (such that they form an independent matrix) in the nonbasic part in $\mathbf{x}$ to make it a BFS for the original problem.

Procedure of the Two-Phase Method

Phase I:

1. Construct the auxiliary problem such that $\mathbf{b} \geq 0$
2. Solve the auxiliary problem using the Simplex method
   - If we reach an optimal solution with optimal value greater than 0, then the original problem is infeasible
3. If the optimal value is 0 with optimal solution $\mathbf{x}^*$, then we enter phase II

Phase II: Solve the original problem starting from the BFS $\mathbf{x}^*$

- If $\mathbf{x}^*$ is degenerate, then we need to supplement some indices to make it a BFS for the original problem

The Big-M method

There is another method that can be used to solve LP without a starting BFS. Consider the following auxiliary problem:

$$\begin{aligned} \text{minimize} \quad & \mathbf{c}^T\mathbf{x} + M\sum_{i=1}^{m} y_i \\ \text{subject to} \quad & A\mathbf{x} + \mathbf{y} = \mathbf{b} \\ & \mathbf{x}, \mathbf{y} \geq 0 \end{aligned}$$

This problem has an initial BFS $\mathbf{y} = \mathbf{b} \geq 0$ (again assuming $\mathbf{b} \geq 0$). Now we can use simplex to solve it. In the simplex procedure, pretend that M is a very large value (larger than any specified number).

- If the original problem is feasible, then optimal $\mathbf{y}$ must be 0
- Two-Phase is more common

Simplex Tableau

Now we have obtained the algebraic procedures for the simplex method.

- We want to have a simpler implementation of it — in particular, we want to avoid explicit matrix inversion in the calculation

We are going to introduce the simplex tableau, which is a practical way to implement the simplex method.

- The simplex tableau maintains a table of numbers
- It visualizes the procedures of the simplex algorithm and facilitates the computation
- After learning the simplex tableau, one should be able to solve small-sized linear optimization problems by hand

We will introduce the simplex tableau in the next lecture.